

Deploying a Java Web Application Using Maven, Nexus, and Tomcat in a Multi-Server Environment

Introduction

In this project, I implemented a complete multi-server Java application deployment pipeline using:

- GitHub as Source Code Repository
- Apache Maven for Build Automation
- Nexus Repository Manager for Artifact Storage
- Apache Tomcat for Application Deployment

The goal of this Proof of Concept (POC) was to simulate a real-world DevOps deployment workflow where applications are built, stored, and deployed across separate servers.

Prerequisites

Before starting, ensure the following:

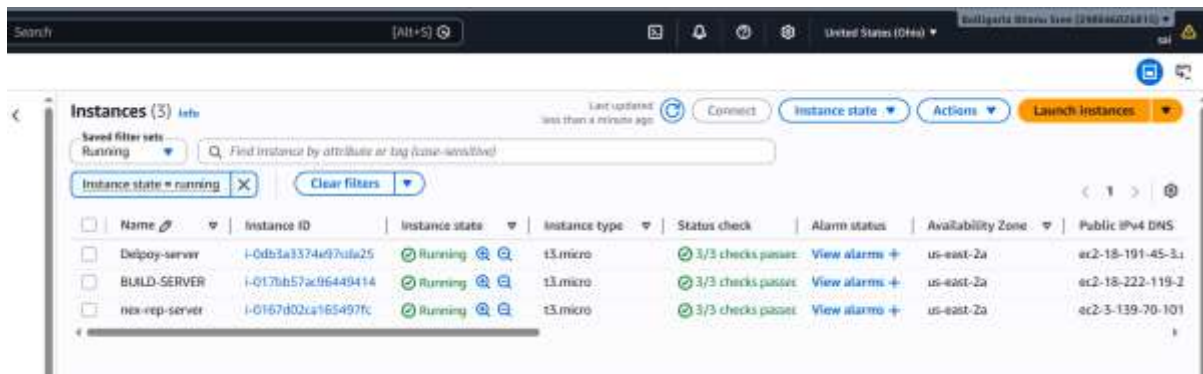
- 3 Ubuntu Servers
- Internet Connectivity
- Open Ports:
 - 22 → SSH
 - 8081 → Nexus
 - 8080 → Tomcat
- Java Installed on All Servers

Step 1 — Configure Build Server

Objective

The Build Server is responsible for:

- Cloning source code
- Building the application
- Uploading the WAR artifact to Nexus



1.2 Install Java

`sudo apt install openjdk-17-jdk -y`

```
See 'git help git' for an overview of the system.
ubuntu@BUILD-SERVER:~$ sudo apt install openjdk-17-jdk
Installing:
  openjdk-17-jdk

Installing dependencies:
  adwaita-icon-theme  libavahi-common3  libjpeg-turbo8  libxdamage1
  alsa-topology-conf  libc-dev-bin      libjpeg8        libxdmcp-dev
```

1.3 Install Maven

`sudo apt install maven -y`

```
No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@BUILD-SERVER:~$ mvn
Command 'mvn' not found, but can be installed with:
sudo apt install maven
ubuntu@BUILD-SERVER:~$ sudo apt install maven
Installing:
  maven
```

1.4 Install Git

`sudo apt install git -y`

Clone GitHub Repository

```
No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@BUILD-SERVER:~$ mvn
Command 'mvn' not found, but can be installed with:
sudo apt install maven
ubuntu@BUILD-SERVER:~$ sudo apt install maven
Installing:
  maven
```

1.5 Build the Application

Run Maven Build

mvn clean install

```
jar (262 kB at 1.1 MB/s)
[INFO] Building jar: /home/ubuntu/JavaCalculator/target/RaviCalculator-1.3.jar
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 9.985 s
[INFO] Finished at: 2026-05-15T10:29:14Z
[INFO] -----
ubuntu@BUILD-SERVER:~/JavaCalculator$ ls
pom.xml  src  target
ubuntu@BUILD-SERVER:~/JavaCalculator$ cd target
ubuntu@BUILD-SERVER:~/JavaCalculator/target$ ls
RaviCalculator-1.3.jar  generated-sources  maven-archiver  surefire-reports
classes                generated-test-sources  maven-status    test-classes
ubuntu@BUILD-SERVER:~/JavaCalculator/target$
ubuntu@BUILD-SERVER:~/JavaCalculator/target$ |
```

Step 2 — Configure Nexus Repository Server

Objective

The Nexus Server stores application artifacts generated by the Build Server.

2.1 Install Java

```
sudo apt update -y
sudo apt install openjdk-17-jdk -y
```

2.2 Download Nexus

Download: `wget https://download.sonatype.com/nexus/3/latest-unix.tar.gz`

Extract: `tar -xvf latest-unix.tar.gz`

2.3 Create Nexus User

```
sudo adduser nexus
```

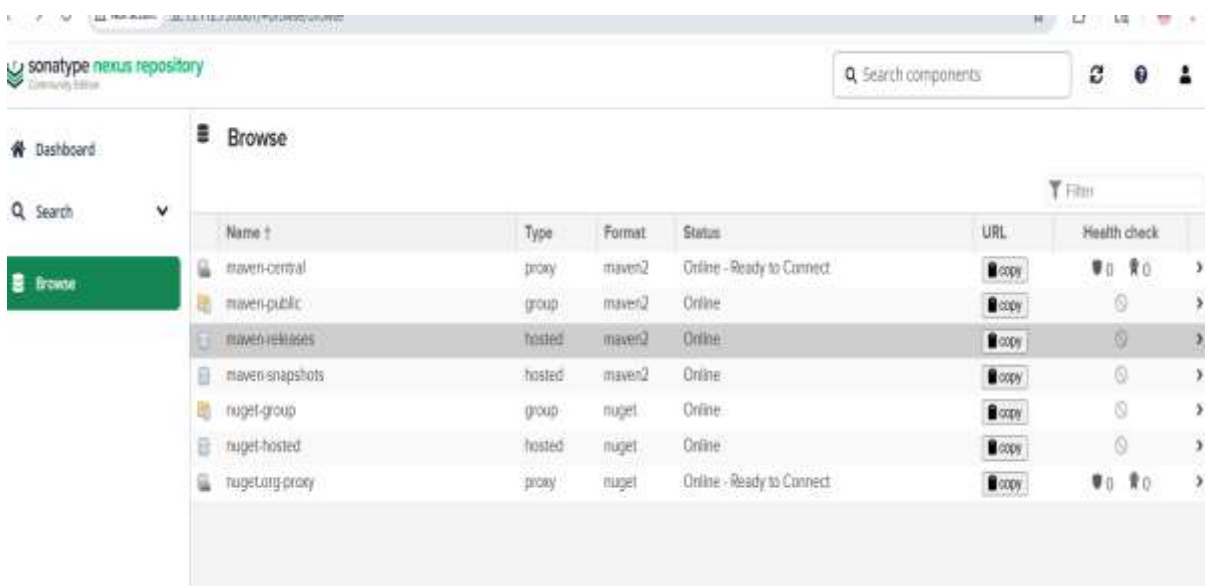
```

ubuntu@nexusREPO:~$ ls
nexus-3.85.0-03  nexus-3.85.0-03-linux-x86_64.tar.gz  sonatype-work
ubuntu@nexusREPO:~$ cd nexus-3.85.0-03/
ubuntu@nexusREPO:~/nexus-3.85.0-03$ ls
NOTICE.txt  OSS-LICENSE.txt  bin  deploy  etc  jdk
ubuntu@nexusREPO:~/nexus-3.85.0-03$ cd bin
ubuntu@nexusREPO:~/nexus-3.85.0-03/bin$ ls
nexus  nexus.vmoptions  sonatype-nexus-repository-3.85.0-03.jar
ubuntu@nexusREPO:~/nexus-3.85.0-03/bin$ ./nexus
Usage: ./nexus {start|stop|run|run-redirect|status|restart|force-reload}
ubuntu@nexusREPO:~/nexus-3.85.0-03/bin$ ls
nexus  nexus.vmoptions  sonatype-nexus-repository-3.85.0-03.jar
ubuntu@nexusREPO:~/nexus-3.85.0-03/bin$ ./nexus start
Starting nexus
ubuntu@nexusREPO:~/nexus-3.85.0-03/bin$ sudo apt update
sudo apt install net-tools -y
Hit:1 http://us-east-2-ec2.archive.ubuntu.com/ubuntu InRelease

```

2.4 Access Nexus UI

Browser: http://<NEXUS_SERVER_IP>:8081



Step 3 — Configure Maven to Upload Artifact to Nexus

Objective

Configure Maven so that WAR files are automatically uploaded to Nexus.

3.1 Update pom.xml

```
        <!-- give full qualified name of your main class-->
        <mainClass>com.ravi.cal.RaviCalculator.Calculator</mainClass>
    </manifest>
</archive>
</configuration>
</plugin>
</plugins>
</build>
<distributionManagement>
    <repository>
        <id>maven-releases</id>
        <url>http://52.15.112.73:8081/repository/maven-releases/</url>
    </repository>
</distributionManagement>
</project>

ubuntu@BUILD-SERVER:~/JavaCalculator$ |
```

3.2 Configure Maven settings.xml

```
        <id>deploymentRepo</id>
        <username>repouser</username>
        <password>repopwd</password>
    </server>
    -->
<servers>
    <server>
        <id>maven-releases</id>
        <username>admin</username>
        <password>admin</password>
    </server>
</servers>

    <!-- Another sample, using keys to authenticate.
    <server>
```

3.3 Deploy Artifact to Nexus

Run: mvn clean deploy

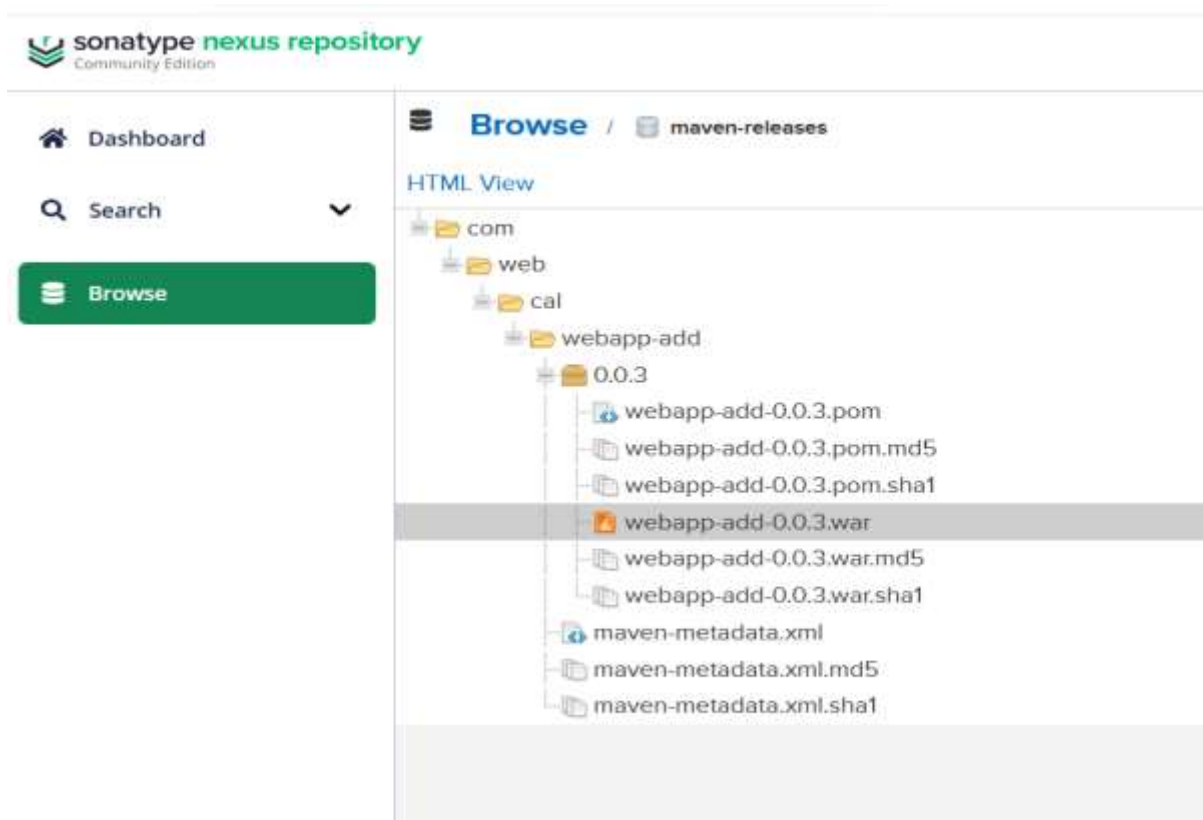
Expected Output: BUILD SUCCESS

3.4 Verify Artifact in Nexus

Navigate:

Browse → maven-releases

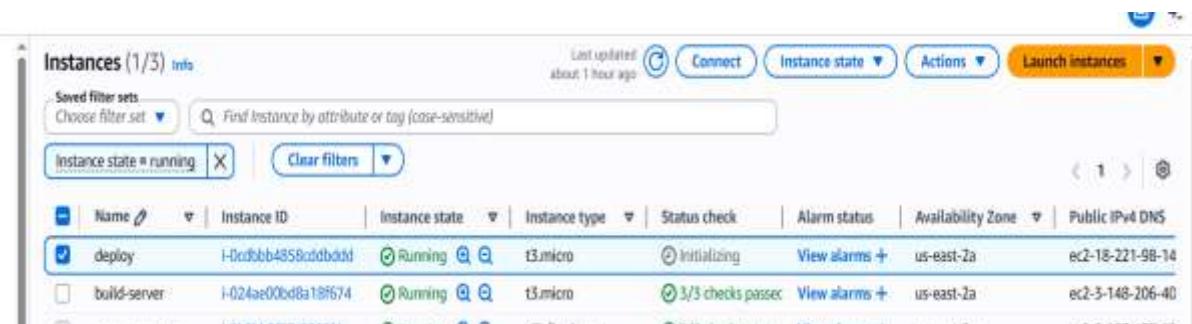
Verify WAR artifact uploaded successfully.



Step 4 — Configure Deploy Server

Objective

The Deploy Server downloads the WAR file from Nexus and deploys it into Apache Tomcat.



4.1 Install Java

```
sudo apt update -y
sudo apt install openjdk-17-jdk -y
```

```
--<name> <value>.  
ubuntu@deploy:~$ java --version  
openjdk 17.0.18 2026-01-20  
OpenJDK Runtime Environment (build 17.0.18+8-Ubuntu-1)  
OpenJDK 64-Bit Server VM (build 17.0.18+8-Ubuntu-1, mixed mode, sharing)  
ubuntu@deploy:~$ |
```

4.2 Install Tomcat

Official Website: `wget https://downloads.apache.org/tomcat/tomcat-9/v9.0.118/bin/apache-tomcat-9.0.118.tar.gz`

Extract: `tar -xvf apache-tomcat-9.0.118.tar.gz`

4.3 Start Tomcat

```
cd /opt/tomcat/bin  
sudo sh startup.sh
```

[Home](#) [Documentation](#) [Configuration](#) [Examples](#) [Wiki](#) [Mailing Lists](#) [Find Help](#)

Apache Tomcat/9.0.118



If you're seeing this, you've successfully installed Tomcat. Congratulations!



Recommended Reading:

- [Security Considerations How-To](#)
- [Manager Application How-To](#)
- [Clustering/Session Replication How-To](#)

Server Status

Manager App

Host Manager

Developer Quick Start

Tomcat Setup	Realms & AAA	Examples	Servlet Specifications
First Web Application	JDBC DataSources		Tomcat Versions

Managing Tomcat

For security, access to the

Documentation

Tomcat 9.0 Documentation

Getting Help

FAQ and Mailing Lists

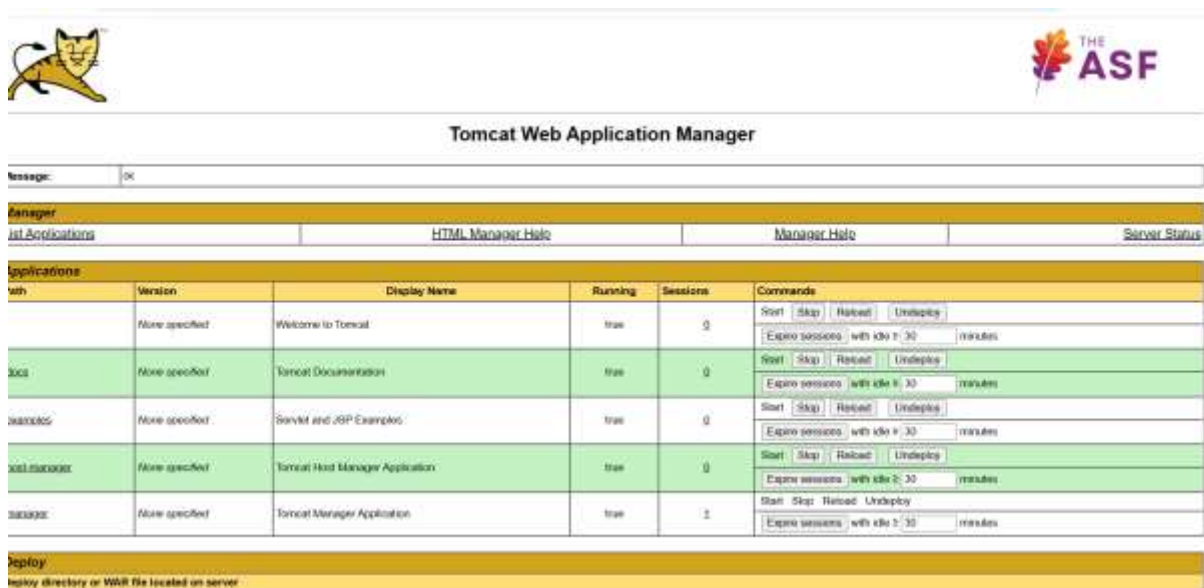
Step 5 — Configure Tomcat Manager Access

5.1 Update tomcat-users.xml

Edit: `sudo vi /opt/tomcat/conf/tomcat-users.xml`

5.2 Allow Remote Access

Edit: `sudo vi /opt/tomcat/webapps/manager/META-INF/context.xml`



Step 6 — Download WAR File from Nexus

6.1 Download Artifact

```
buntu@deploy:~/apache-tomcat-9.0.118$ cd webapps/
buntu@deploy:~/apache-tomcat-9.0.118/webapps$ ls
DOT docs examples host-manager manager
buntu@deploy:~/apache-tomcat-9.0.118/webapps$ wget http://3.137.177.89:8081/repository/maven-releases/com/web/cal/webapp-add/0.0.3/webapp-add-0.0.3.war
--2026-05-18 12:58:12-- http://3.137.177.89:8081/repository/maven-releases/com/web/cal/webapp-add/0.0.3/webapp-add-0.0.3.war
Connecting to 3.137.177.89:8081... connected.
HTTP request sent, awaiting response... 200 OK
Length: 3981 (3.9K) [application/java-archive]
```

Step 7 — Deploy WAR to Tomcat

Copy WAR: `cp calculator-1.0.war /opt/tomcat/webapps/`

Step 8 — Access Application

Open Browser: Application will host.

Calculator

first number:

Second number :

☐ addition

☐ subtraction

☐ production

Challenges Faced

Issue 1 - Tomcat Manager Access Denied

Solution

- Added roles in tomcat-users.xml
- Commented RemoteAddrValve
- Restarted Tomcat

Issue 2 - Nexus Authentication Failure

Solution

- Corrected credentials in settings.xml
- Verified repository permissions

Issue 3 - WAR Not Deploying

Solution

- Checked WAR file location
- Restarted Tomcat

- Verified Java compatibility

Conclusion

This project successfully demonstrated a real-world multi-server Java deployment pipeline using Maven, Nexus, and Apache Tomcat.